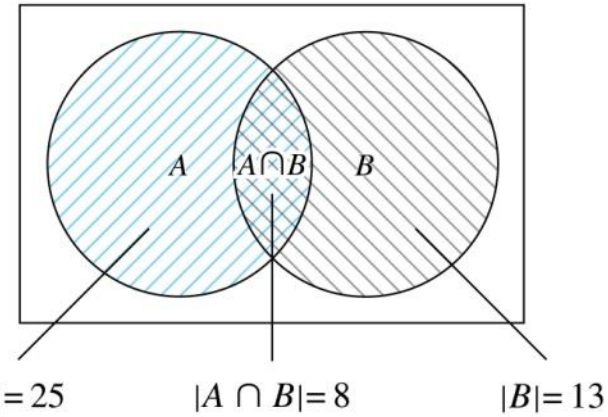
The Inclusion-Exclusion Principle

- Reading
  - Rosen, 8.5, 8.6

Cardinality of Union: Example 1

- In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

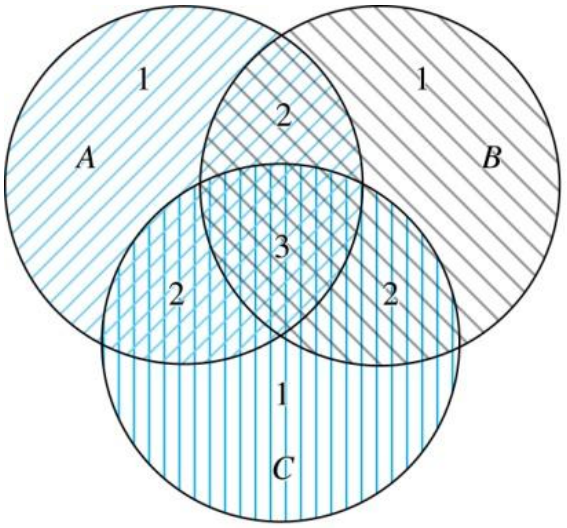
$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$$



Three Finite Sets

$$|A \cup B \cup C| =$$

$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

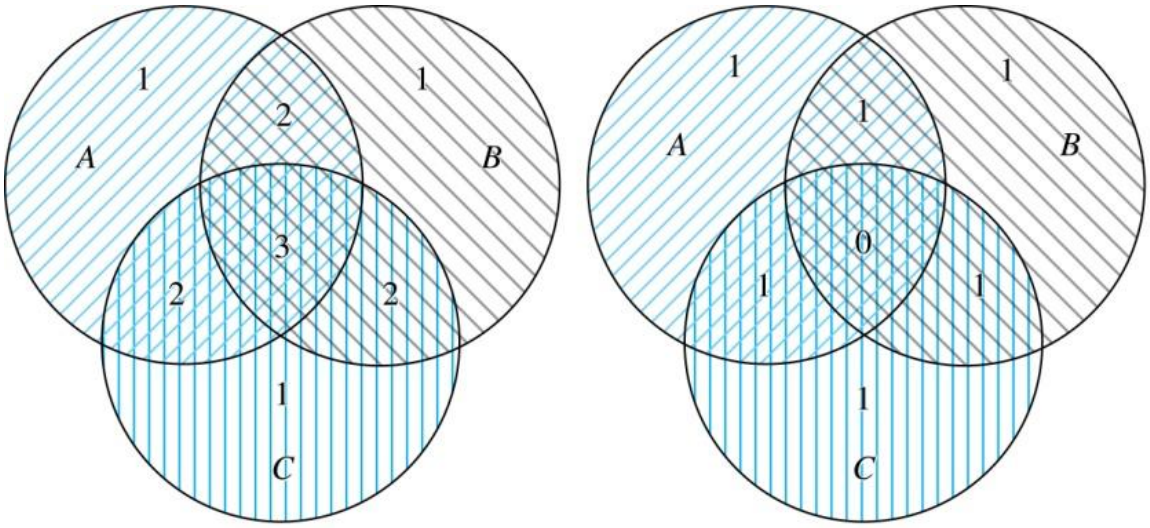


(a) Count of elements by $|A|+|B|+|C|$

Three Finite Sets

$$|A \cup B \cup C| =$$

$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

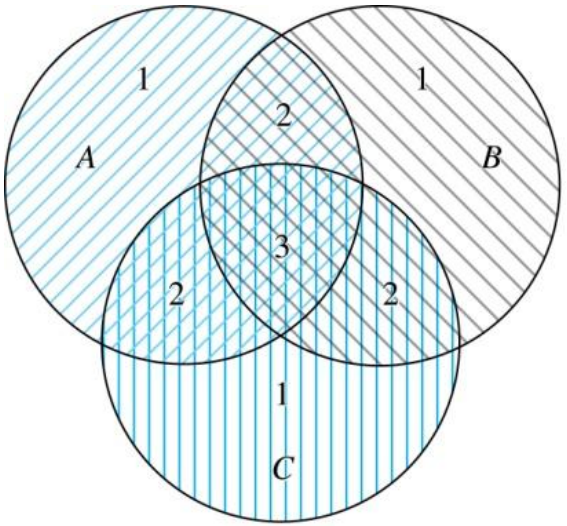


(a) Count of elements by $|A|+|B|+|C|$

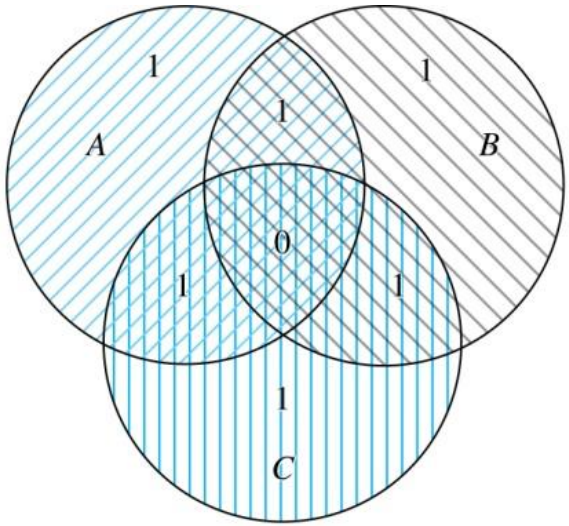
(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$

Three Finite Sets

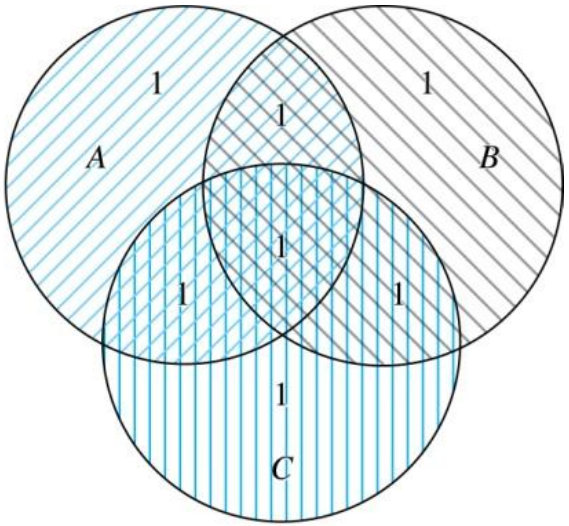
$$|A \cup B \cup C| =$$
$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



(a) Count of elements by $|A|+|B|+|C|$



(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$



(c) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$

Example with 3 Finite Sets

- 1232 students have taken a course in Spanish;
- 879 have taken a course in French;
- 114 have taken a course in Russian;
- 103 have taken courses in both Spanish and French;
- 23 have taken courses in both Spanish and Russian;
- 14 have taken courses in both French and Russian;
- 2092 students have taken a course in at least one of Spanish French and Russian.
- How many students have taken all 3 languages?
- Solutions:

  $|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|$,

  we obtain $2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$.

  Solving for $|S \cap F \cap R|$ yields 7.

Inclusion-Exclusion Principle

- **Theorem**
  Let $A_1, A_2, \ldots, A_n$ be finite sets. Then:

$$|A_1 \cup A_2 \cup \cdots \cup A_n| =$$

$$\sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| +$$

$$\sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots + (-1)^{n+1} |A_1 \cap A_2 \cap \ldots \cap A_n|$$

Proof

- We will prove the formula by showing that an element in the union is counted exactly once by the right-hand side of the equation.
- Suppose a is a member of exactly r ($1 \le r \le n$) of the sets
  - Example: a belongs to $r = 3$ sets A_1,A_2 and A_3)
- This element is counted:
- $C(r,1)$ times by $\sum|A_i|$
  - if $r = 3$, $C(3,1) = 3$ times in A_1,A_2 and A_3
- $C(r,2)$ times by $\sum|A_i \cap A_j|$
  - if $r = 3$, $C(3,2) = 3$ times in $A_1 \cap A_2, A_1 \cap A_3$ and $A_2 \cap A_3$
- $C(r,3)$ times by $\sum|A_i \cap A_j \cap A_j|$
  - if $r = 3$, $C(3,3) = 1$ time in $A_1 \cap A_2 \cap A_3$
- In general, the element is counted $C(r,m)$ times by each sum involving m sets.
- Thus, a is counted exactly

$$C(r,1) - C(r,2) + C(r,3) - \cdots + (-1)^{r+1}C(r,r)$$

times. We must prove that this is equal to 1

Proof (cnt'd)

- Recall from previous lecture that a corollary of the binomial theorem states (the number of subsets that have even cardinality equals the number of subsets that have odd cardinality):

$$\sum_{k=0}^{n} \binom{n}{k}(-1)^k = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \binom{n}{4} - \binom{n}{5} + \cdots = 0$$

Thus, setting $n = r$:

$$C(r,0) - C(r,1) + C(r,2) - C(r,3) + \cdots + (-1)^r C(r,r) = 0.$$

- Hence,

$$C(r,0) = 1 = C(r,1) - C(r,2) + C(r,3) - \cdots + (-1)^{r+1} C(r,r)$$

Exercise on Employees and Tasks

- How many ways are there to assign 6 different tasks to 3 different employees if every employee is assigned at least one task?
- How many assignments would be possible if there were no constraint?
- $3^6=729$, 3 options for 1st task, 3 options for 2nd,…(permutations with repetition)
- Wrong Solution for constrained version:

1. Step 1: Choose the first 3 tasks (C(6,3)) and assign them to 3 different employees: $C(6,3)\cdot 3!=P(6,3)$. Each employee is assigned a task
2. Step 2: Assign the remaining 3 tasks to any employee: 3^3

- Total number of ways: $C(6,3)\cdot 3!\cdot 3^3=20\cdot 6\cdot 27=3240$
- Mistake: Overcounting. The correct solution should be less than 729.
- Assume that in an assignment x, Step 1 one assigns task 1 to employee 1 and Step 2 assigns task 2 to employee 1.
- Assume another assignment x', which is identical to x (all tasks are assigned to the same employees), but task 2 is assigned to employee 1 at Step 1, and task 1 is assigned to employee 1 at Step 2.
- x and x' are counted as distinct solutions, whereas they are equivalent.

Number of Onto Functions (special case)

- Exercise: How many ways are there to assign 6 different tasks to 3 different employees if every employee is assigned at least one job?
- Equivalent Problem: How many surjective (or onto) functions are there from a set A with 6 elements to a set B with 3 elements?
- Correct Solution
- What is the total number of functions (without any constraint) from A to B?
- Every element of A is mapped to any element in B: 3^6=729. But I have to remove non-surjective functions.

Let $B = \{b_1, b_2, b_3\}$. Let P_i be the set of functions that do not have $b_i$in their range; e.g, P_1 is the set of functions that do not map to b_1. The functions of P_i are not onto and should be removed from the 3^6 solutions.

- What is the value of $|P_1|$, $|P_2|$, $|P_3|$?
- $|P_1| = |P_2| = |P_3| = 2^6$ - There are 2 choices to map each element of A.
- Is the number of onto functions $3^6 - 3 \cdot 2^6$ =537?
- No. Some functions in P_1 may also belong to P_2 or P_3. I need to use inclusion-exclusion, to compute the number of non-surjective functions $|P_1 \cup P_2 \cup P_3|$.

Number of Onto Functions(special case-cont)

- # of functions P_i that do not have b_i in their range: $|P_1| = |P_2| = |P_3| = 2^6$
- I need to use inclusion-exclusion, to compute $|P_1 \cup P_2 \cup P_3|$. Inclusion-exclusion for 3 sets: $|P_1 \cup P_2 \cup P_3| = |P_1| + |P_2| + |P_3| - |P_1 \cap P_2| - |P_2 \cap P_3| - |P_1 \cap P_3| + |P_1 \cap P_2 \cap P_3|$
- Give an example of a function that belongs to $(P_1 \cap P_2)$
- A function that maps all elements of A to b_3.
- What is the value of $|P_1 \cap P_2|, |P_2 \cap P_3|, |P_1 \cap P_3|$?
- $|P_1 \cap P_2| = |P_2 \cap P_3| = |P_1 \cap P_3| = 1$ - There is 1 choice for each element of A.
- What is the value of $|P_1 \cap P_2 \cap P_3|$?
- $|P_1 \cap P_2 \cap P_3| = 0$ - There is 0 choice for each element of A.
- Thus, $|P_1 \cup P_2 \cup P_3| = 3 \cdot 2^6 - 3 \cdot 1 = 189$ non-surjective functions.
- The number of surjective functions is 729-189=540

Number of Onto Functions (general case)

- How many surjective (or onto) functions are there from a set A with m elements to a set B with n elements ($m \geq n$)?
- Solution
- What is the total number of functions (without any constraint) from A to B?
- n^m
- Let P_i be the set of functions that do not have b_i in their range.
- What is the value of each $|P_i|$?
- $(n-1)^m$ because each element of A can be mapped to every element of B except of b_i.
- Since the sets P_i may contain common elements, I use inclusion-exclusion to compute the number of non-surjective functions $|P_1 \cup P_2 \cup \cdots \cup P_n|$. The number of surjective functions is $n^m - |P_1 \cup P_2 \cup \cdots \cup P_n|$.
- $|P_1 \cup P_2 \cup \cdots \cup P_n| =$

$$\sum_{1 \leq i \leq n} |P_i| - \sum_{1 \leq i < j \leq n} |P_i \cap P_j| + \sum_{1 \leq i < j < k \leq n} |P_i \cap P_j \cap P_k| - .... + (-1)^{n+1} |P_1 \cap P_2 \cap ... \cap P_n|$$

Number of Onto Functions (general case-cont)

$$\sum_{1\le i\le n} |P_i| - \sum_{1\le i<j\le n} |P_i \cap P_j| + \sum_{1\le i<j<k\le n} |P_i \cap P_j \cap P_k| - \ldots + (-1)^{n+1} |P_1 \cap P_2 \cap \ldots \cap P_n|$$

- What is the value of each $|P_i \cap P_j|$?
- $(n\text{-}2)^m$ because each element of A can be mapped to every element of B except of b_i and b_j.
- What is the value of $|P_1 \cap P_2 \cap \cdots \cap P_r|$ (and every intersection of r sets)?
- $(n\text{-}r)^m$ because each element of A can be mapped to every element of B except of b_1, b_2 … b_r.
- What is the value of $|P_1 \cap P_2 \cap \cdots \cap P_n|$
- 0
- How many intersections of r sets are there?
- C(n,r)
- Thus, the above equation becomes: $|P_1 \cup P_2 \cup \cdots \cup P_n| =$

$$C(n,1)(n-1)^m - C(n,2)(n-2)^m + C(n,3)(n-3)^m + .. + (-1)^n C(n,n-1)1^m$$

Derangement

- A **derangement** is a permutation of objects that leaves no object in its original position.
- Example
  The permutation 21453 is a derangement of 12345 because no number is left in its original position. However, 21543 is not a derangement of 12345 because this permutation leaves 4 unchanged.
- Theorem
  The number of derangements of a set with n elements is

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n \frac{1}{n!}\right]$$

Number of Permutations that are not Derangements

- Assuming permutations of n numbers 1,2,..,i,..,n, we will count the number of permutations that are not derangements.
- Let P_i be the set of permutations, where i remains in its initial position (these permutations are not derangements).
- What should I compute?
- $|P_1 \cup P_2 \cup \ldots \cup P_i \cup \ldots \cup P_n|$ by inclusion - exclusion
- What is the value of $|P_i|$?
- $(n-1)!$ since i must remain in the same position, I can only permute the remaining n-1 numbers.
- Which permutations belong to $P_i \cap P_j$?
- Permutations that leave i and j in their original positions.
- What is the value of $|P_i \cap P_j|$?
- $(n-2)!$
- What is the value of $|P_1 \cap P_2 \cap \ldots \cap P_r|$ and every intersection of r sets?
- $(n-r)!$
- How many intersections of r sets are there?
- $C(n,r)$

Number of Permutations that are not Derangements - 2

- Assuming permutations of *n* numbers 1,2,..,*i*,..,n, we will count the number of permutations $|P_1 \cup P_2 \cup \ldots \cup P_n|$ that are not derangements, by inclusion - exclusion.
- $|P_1 \cup P_2 \cup \cdots \cup P_n| =$

$$\sum_{1 \le i \le n} |P_i| - \sum_{1 \le i < j \le n} |P_i \cap P_j| + \sum_{1 \le i < j < k \le n} |P_i \cap P_j \cap P_k| - \ldots + (-1)^{n+1} |P_1 \cap P_2 \cap \ldots \cap P_n|$$

$$= \binom{n}{1}(n-1)! - \binom{n}{2}(n-2)! + \cdots + (-1)^{n+1}\binom{n}{n}(n-n)!$$

$$= \frac{n!}{1!\,(n-1)!}(n-1)! - \frac{n!}{2!\,(n-2)!}(n-2)! + \cdots + (-1)^{n+1}\frac{n!}{n!\,0!}0!$$

$$= n!\left[\frac{1}{1!} - \frac{1}{2!} + \cdots + (-1)^{n+1}\frac{1}{n!}\right]$$

Number of derangements:

$$D_n = n! - n!\left[\frac{1}{1!} - \frac{1}{2!} + \cdots + (-1)^{n+1}\frac{1}{n!}\right] = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n \frac{1}{n!}\right]$$

Exercise on Derangements

- A machine that inserts letters into envelopes goes haywire and inserts letters randomly into envelopes. In a group of 100 letters, how many ways if

a) No letter is put into the correct envelope?

Answer: D_{100}

b) Exactly 1 letter is put into the correct envelope?

Answer: C(100,1) $\cdot D_{99}$ ways to choose the letter correctly stuffed. D_{99} ways to insert the remaining 99 letters all incorrectly.

c) Exactly 98 letters are put into the correct envelopes?

Answer: C(100,98)=C(100,2). Choose two letters to be misplaced and there is only one way to misplace them.

Exercise from previous class

- How many solutions does the equation $x_1 + x_2 + x_3 = 5$ have, where x_1, x_2, and x_3 are nonnegative integers? Equivalently, in how many way I can select 5 cookies of 3 different kinds (*combinations with repetition*).
- $C(3 + 5 - 1,5) = C(7,5) = 21$
- How many solutions does the equation $x_1 + x_2 + x_3 = 5$ have, where x_1, x_2, and x_3 are nonnegative integers and $x_3 \geq 3$?
- I rewrite the equation as follows:
- $x_1 + x_2 + (x_3 - 3) = 5 - 3 \Rightarrow x_1 + x_2 + x = 2$.
- The number of solutions is: $C(3 + 2 - 1,2) = C(4,2) = 6$
- How many solutions does the equation $x_1 + x_2 + x_3 = 5$ have, where x_1, x_2, and x_3 are nonnegative integers and $x_3 \geq 3$ and $x_1 \geq 1$?
- Using the previous reasoning, I rewrite $(x_1 - 1) + x_2 + x = 2 - 1$.
- The number of solutions is: $C(3 + 1 - 1,1) = C(3,1) = 3$

Exercise on Inclusion-Exclusion

- How many solutions does the equation $x_1 + x_2 + x_3 = 5$ have, where x_1, x_2, and x_3 are nonnegative integers and x_3<3 and x_1<1?
- You can use the following results

solutions without constraint: $|S_{noCon}| = C(3+5-1,5) = C(7,5) = 21$

solutions where $x_3 \geq 3$: $|S_{x3\geq 3}| = C(4,2) = 6$

solutions where $x_1 \geq 1$: $|S_{x1\geq 1}| = C(6,4) = 15$

solutions where $x_3 \geq 3 \wedge x_1 \geq 1$: $|S_{x3\geq 3 \text{ and } x1\geq 1}| = C(3,1) = 3$

Answer (by inclusion exclusion):

- From the total number of solutions, remove those $x_3 \geq 3$ and $x_1 \geq 1$, and add those with $x_3 \geq 3 \wedge x_1 \geq 1$
- $C(7,5)$-$C(4,2)$- $C(6,4)$ + $C(3,1)$=21-6-15+3=3

Since x_1<1, $x_2 + x_3 = 5$. Since x_3<3, the three solutions of $x_2 + x_3 = 5$ are $x_2 = 5, x_3 = 0$, or $x_2 = 4, x_3 = 1$, or $x_2 = 3, x_3 = 2$.

Exercise on Employees and Tasks

- How many ways are there to assign 5 distinct tasks to 4 different people such that every person has at least one task assigned?
- Solution 1: This is the number of surjective functions from a set of 5 elements to a set of 4 elements: 4^5-C(4,1)3^5 + C(4,2)2^5 - C(4,3)1^5 =1024-972+192-4 = 240
- Solution 2: A person must be assigned 2 tasks, and the rest 1 task each. Select 2 tasks and assign them to some person in C(5,2)x4=40 ways. Then assign the remaining tasks to 3 persons in 3!=6 ways. Total is C(5,2)x4 x3!=240
- Solution 3: Select 4 tasks and assign them to 4 persons in C(5,4)x4!=5x24=120 ways. Then, we assign the remaining task to one of the 4 users in 4 ways. This solution (C(5,4)x4!x4=480) double counts for the user that has 2 tasks (e.g., (task1, task2) and (task2,task1) are counted as 2 assignments). To avoid this problem we divide by 2, to get the answer C(5,4)x4!x2=240.

Exercise on Employees and Tasks 2

- Assume that the 5 tasks are identical. How many ways are there to assign 5 indistinguishable tasks to 4 different people such that every person has at least one task assigned?
- Solution 1: Since all 5 tasks are identical, and each person will be assigned at least 1, you only need to select the person that will be assigned 2 tasks in C(4,1)=4 ways.
- Solution 2: You can assume that each task corresponds to a cookie and you want to distribute them to 4 persons, so that each person receives at least one cookie.
- Equivalent equation: $x_1 + x_2 + x_3 + x_4 = 5$ with $x_1 \geq 1, x_2 \geq 1, x_3 \geq 1, x_4 \geq 1$.
- $x_1 + x_2 + x_3 + x_4 = 1$ in unconstrained version.
- C(4+1-1,1)=C(4,1)=4